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Original Article

Bridging the gap between legacy polymerase chain reaction-based microsatellite data with high-throughput sequencing data for conservation genomics

Dalya Salih[1,2], Ellie E. Armstrong[3,4], Charles T. Robbins[5], Lisette P. Waits[6], Joanna L. Kelley[2,3,*]

[1]Department of Biomolecular Engineering, University of California, Santa Cruz, 1156 High Street, Santa Cruz, CA 95064, United States, [2]Department of Ecology and Evolutionary Biology, University of California, Santa Cruz, 130 McAllister Way, Santa Cruz, CA 95060, United States, [3]School of Biological Sciences, Washington State University, 431 Heald Hall, Pullman, WA 99164, United States, [4]Department of Evolution, Ecology & Organismal Biology, University of California, Riverside, 900 University Ave, Riverside, CA 92521, United States, [5]School of the Environment, Washington State University, Pullman, WA 99164, United States, [6]Department of Fish and Wildlife Sciences, College of Natural Resources, University of Idaho, Moscow, ID 83844, United States

*Corresponding author: Department of Ecology and Evolutionary Biology, University of California, Santa Cruz, 130 McAllister Way, Santa Cruz, CA 95060, USA. Email: jokelley@ucsc.edu

Abstract

Microsatellites are powerful markers for tracking genetic variation in wildlife populations due to their high polymorphism and genome-wide abundance. While polymerase chain reaction (PCR)-based fragment size analysis has been the standard for genotyping microsatellites, high-throughput sequencing offers greater resolution and the opportunity to sync historical datasets with modern analyses. We evaluated how genotypes from whole-genome sequencing align with PCR data for 15 microsatellite loci in 11 North American brown bears (*Ursus arctos*). Brown bear populations in the 48 contiguous United States have declined from approximately 50,000 to fewer than 2,000 over the past decades. Their endangered status has prompted extensive research and genetic monitoring, yielding large, multiyear microsatellite datasets upon which future conservation efforts can build. We achieved an overall microsatellite genotype concordance rate of 94.5% comparing high-throughput sequencing results to PCR based-fragment size results. All discrepancies occurred at complex loci containing multiple insertions and/or deletions (indels). Physically linked indels or single nucleotide polymorphisms (SNPs) occurring within the loci were misinterpreted as independent insertions, underscoring the need for genotyping tools that incorporate phasing when genotyping. To evaluate coverage effects, we downsampled high-throughput sequence data from 30x to 2x. Concordance remained high at 20 to 30x but dropped sharply at 10x, with 5x and 2x having discordant genotypes or insufficient coverage for genotyping. Accurate genotyping required both sufficient depth and number of reads spanning the entire repeat regions. Our results show that short-read whole-genome sequencing can recover microsatellite genotypes with high accuracy when paired with careful variant interpretation. By aligning historical PCR datasets with modern sequencing data, we can preserve decades of genetic insight and strengthen long-term monitoring of at-risk populations.

Key words: brown bear, concordance, conservation, coverage, genotyping, short tandem repeat

Introduction

Conservation decisions often rely on genetic information in addition to species distributions and habitat suitability (Abdul-Muneer 2014; DeMay et al. 2017). Microsatellites, also known as short tandem repeats (STRs) or simple sequence repeats (SSR), are sequences of 1 to 10 base pairs (bp) that repeat in succession at specific locations throughout the genome (Vieira et al. 2016). These regions predominantly reside in intergenic regions and are valuable markers for assessing genetic variability, population structure, and relatedness. Their variability arises from strand-slippage during DNA replication or recombination errors, which generates many alleles per locus, making them informative for detecting fine-scale genetic differences (Vieira et al. 2016). Due to microsatellites' codominant inheritance, high mutation rate, and abundance across the genome, microsatellites have been widely used over the past decades for genotyping individuals (Allendorf 2025; Storfer et al. 2007). Such genetic data are critical for identifying populations at risk of inbreeding depression or isolation and for supporting decisions regarding translocation and habitat connectivity (Buono et al. 2022). Microsatellite loci have also been the main marker type used for genetic monitoring of wildlife (Schwartz et al. 2007; Stetz et al. 2011).

Traditionally, polymerase chain reaction (PCR) has been used to amplify microsatellite regions by targeting known non-repetitive flanking sequences. These PCR products are then separated by size allowing fragment length analysis methods to assess genetic variability and relatedness (Vieira et al. 2016). PCR-based genotyping has been a cornerstone of microsatellite analysis for decades due to its efficiency, accessibility, and ability to generate reliable genotypic data from limited DNA samples. Its sensitivity makes it especially valuable in studies using degraded or low-quantity DNA, such

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as ancient specimens or noninvasive samples (Taberlet et al. 1996).

Many PCR-based microsatellite datasets were developed from samples that cannot be recollected, including museum skins, archival tissues, non-invasive field samples, and even remains from extinct populations. Because PCR can amplify DNA from very small and degraded sources, these datasets capture unique historical records of genetic variation that are otherwise inaccessible (Geffen et al. 2007). As a result, they represent irreplaceable baselines for assessing how populations have changed over time. Reconciling these fragment-based genotypes with modern sequencing data is therefore essential for maintaining continuity in conservation genetics and for extending the value of past sampling into present-day analyses.

However, while widely used, PCR-based microsatellite analysis can suffer from errors such as stutter peaks from PCR slippage, allele dropout when one allele fails to amplify, and scoring inconsistencies between labs and researchers, which can compromise data quality and reproducibility (Taberlet et al. 1997; Guichoux et al. 2011). Workflows for PCR-based genotyping also depend on equipment that has become increasingly less accessible to researchers. In contrast, high-throughput sequencing platforms have become more broadly available and cost-effective, offering scalable solutions for genotyping across many individuals and species (Guichoux et al. 2011).

PCR fragment length analysis only measures the overall size of the amplified fragment, which means that important details like single nucleotide polymorphisms (SNPs) within the repeat region go undetected (Bull et al. 1999; Lepais et al. 2020). Moreover, if there are multiple insertions and deletions (indels) within the region, only the total length will be reported (Bull et al. 1999). However, alleles can appear identical in size but differ in sequence due to independent mutational events. This is referred to as homoplasy. New sequencing technologies now make it possible to capture the exact base-by-base sequence within microsatellite regions, including the repeat motif and any internal SNPs or indels. This allows distinguishing between alleles that may appear identical in length by PCR but actually differ in sequence, giving a clearer view of complex genetic variation (Lepais et al. 2020). This improvement in resolution increases the number of alleles and disambiguating homoplasy, therefore providing more accurate information for genetic monitoring.

Historically, the development of STR markers required labor-intensive cloning and Sanger sequencing, which limited the pace and scale of microsatellite-based genetic research (Abdul-Muneer 2014; Vieira et al. 2016), especially for non-model organisms due to the absence of reference genomes or known sequence data. High-throughput sequencing overcomes this bottleneck by rapidly producing large volumes of sequence data, enabling de novo identification of STR motifs, including in previously uncharacterized genomes (Guichoux et al. 2011). While high-throughput sequencing offers comprehensive, genome-wide insights into mutation rates, structural variants, and population history, it poses its own challenges when applied to microsatellites. Short-read sequences (typically 100 to 300 bp) may not span the full length of longer or highly repetitive microsatellite regions, potentially leading to incomplete or inaccurate repeat length estimates. Sequencing depth, the average number of reads covering each STR locus, influences how confidently sequence-based genotypes can be determined. While increased sequencing depth can improve accuracy, existing research shows that the optimal depth for accurate microsatellite analysis remains unclear (Zavodna et al. 2014).

For decades, PCR-based microsatellite genotyping has generated valuable datasets for population genetics, conservation, forensics, and medicine (Allendorf et al. 2022). As sequencing technologies advance, it is essential to ensure these historical datasets can be used and remain compatible with modern approaches. Ensuring compatibility between old and new data allows modern sequencing to extend the value of previous studies, refining and augmenting, rather than discarding, what PCR-based methods uncovered.

Our goal here was to develop a workflow that effectively integrates PCR-based microsatellite genotyping data with modern high-throughput sequencing data. Across many species, microsatellite datasets have been collected over years often using PCR-based methods (for example in De Barba et al. 2010, DeMay et al. 2017). These datasets offer critical insights into long-term population trends and genetic diversity. However, the transition to high-throughput sequencing has created a disconnect between historical records and modern analyses because fragment-based genotypes are in the format of allele sizes that are not necessarily compatible with DNA sequence-based genotypes. Moreover, fragment-based genotype lengths are often dependent on the particular lab and standard that were used for genotyping. Currently there is no method to link whole-genome sequence data to previously generated microsatellite datasets. Bridging this gap allows us to preserve the value of existing data while taking advantage of the depth and resolution that high-throughput sequencing offers.

Brown bears (*Ursus arctos*) have a long history of microsatellite-based monitoring and conservation genetics. Over the past few decades, microsatellites have revealed genetic diversity, parentage, and population structure, detected bottlenecks, enumerated population size, and guided management in fragmented bear populations across Europe, Asia, and North America (Taberlet et al. 1997; Paetkau et al. 1998; Paetkau and Strobeck 1998; Woods et al. 1999; Mowat et al. 2005; De Barba et al. 2010; Scarpulla et al. 2021; Turnock et al. 2025). To investigate the potential to link previously generated microsatellite data with contemporary high-throughput sequencing studies, we genotyped 15 commonly used microsatellites in 11 brown bears using traditional PCR-based methods and DNA sequencing-based methods with high-coverage, whole-genome sequencing from the same individuals. This side-by-side approach gives us a rare opportunity to examine how previous genetic monitoring methods and modern data types align between the same individuals. High-throughput sequencing data and PCR-based microsatellite data are often collected across different time periods or from different individuals, making it difficult to evaluate how compatible the two approaches are. By evaluating the methods with the same individuals, we calculate the concordance of the two approaches. Moreover, by exploring various sequencing depths, we assess how coverage impacts concordance between the methods, pinpointing the minimum depth needed to accurately call microsatellite genotypes from high-throughput sequencing data.

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Methods

Bear sampling

Animals

Brown bears (*U. arctos*) were housed and sampled at the Washington State University Bear Research, Education and Conservation Center (WSU Bear Center, Pullman, WA, USA) in accordance with the Bear Care and Colony Health Standard Operating Procedures protocol (#6468) and the Blood and Tissue Sampling protocol (#6546) approved by the Washington State Institutional Animal Care and Use Committee. Eleven individuals were included in our study.

DNA extraction and quantification

Blood was collected (~5 mL) from the jugular vein into PAXgene Blood DNA Tubes. DNA was extracted using the PAXgene Blood DNA Kit following the standard protocol for whole blood with no modifications. DNA was quantified with the high sensitivity double-stranded (ds) DNA Assay Kit (Qubit dsDNA HS Assay Kit; Thermo Fisher Scientific, Q32854) on the Qubit 2.0 fluorometer.

High-throughput sequencing

DNA sequencing

To prepare the samples for sequencing, standard PCR-free libraries were constructed using KAPA Hyper PCR-free library reagents (KK8505; KAPA Biosystems). Genomic DNA was sheared to a size range of 200 to 600 bp and then purified using AMPure XP beads. A double size selection step, employing varying ratios of AMPure XP beads, was used to isolate a specific size range of DNA fragments for library preparation. This was followed by end-repair and 3′-adenylation of the DNA fragments before ligating barcoded adapters. The quality of the prepared libraries was assessed through fragment size analysis and qPCR. Finally, the libraries were sequenced using paired-end reads (150 bp) on an Illumina NovaSeq 6000 at the Baylor College of Medicine Human Genome Sequencing Center (Houston, TX, USA). A subset of these samples were sequenced and analyzed in (Wang et al. 2022).

Microsatellites PCR-based typing

DNA extracted from blood was genotyped in duplicate for 11 brown bears at 15 microsatellite loci, G10B, G1D, G10H, G10M, G1A, G10C, G10L, G10U, G10P, G10X, MU11, MU15, MU23, MU50, REN145P07, (Primer sources, Table S1) using fluorescently labeled primers. PCR conditions are described in (De Barba et al. 2010). These loci have been used extensively in brown bear genetic studies and have been described as unlinked, independent, and meeting Hardy–Weinberg expectations (Proctor et al. 2012).

Typing microsatellites in short-read data

Identifying putative microsatellite locations in the brown bear genome

We used multiple lines of evidence to locate putative microsatellite locations in the brown bear genome assembly (GCF_023065955.2) (Armstrong et al. 2022). We first identified the primers used for each microsatellite locus from various publications (Table S1). We then used BLAST version 2.16.0 (Altschul et al. 1990; Camacho et al. 2009) to locate the position of the primers in the reference genome. To do this, we first created a BLAST database of the reference genome using the *makeblastdb* command with the flag "-type nucl". Subsequently, concatenated forward and reverse primer sequences separated by five "Ns" were provided as a query file and the database provided with the "db" flag. We additionally added flags "-outfmt 6" (outputs into tab format), "-word_size 7" (increases specificity of the search, default value is 28), "-evalue 100" (number of hits observed by chance), "-dust no" (turns off repetitive sequence filtering), "-soft_masking false" (does not ignore any masked sequences), and "-max_target_seqs 5" (limits to the top five hits). These parameters were selected to optimize BLAST to identify short sequences that may occur in repetitive regions and are recommendations typically used when trying to find or verify primer sequences. We manually examined the resulting hits to find regions where the forward and reverse hit were within 700 bp of each other (i.e. where the first half of the concatenated sequence aligned very close to the second half). In the case that there was not a hit within this distance, we retained the sequence(s) with the longest hits and the highest percent identities for further investigation. We prepared a bed file with the coordinates of the previously identified BLAST hits and extracted any simple repeats identified in this region and ran an intersect on this using BEDtools version 2.31.0 (Quinlan and Hall 2010). We checked for microsatellite repeats in the region with repeat motifs of between 2 and 6 nucleotides. Though the microsatellite motifs were previously identified as dinucleotides, we looked for motifs of up to six bp in length to ensure that the repeats were not misidentified previously or had changed over time.

Typing microsatellites with Genome Analysis Toolkit HaplotypeCaller

To genotype microsatellites in the short-read whole genome sequencing data, we used the Genome Analysis Toolkit (GATK) (McKenna et al. 2010; Poplin et al. 2018) version 4.4.0.0 pipeline. The pipeline consists of three tools—HaplotypeCaller, CombineGVCFs, and GenotypeGVCFs. GATK HaplotypeCaller, the first step in the pipeline uses a local de novo assembly strategy to construct candidate haplotypes in regions showing potential variation. HaplotypeCaller captures small indels and repeat expansions.

HaplotypeCaller was run on all samples using the default settings with two additional arguments: we set "–max-reads-per-alignment-start" to 0 to disable the read filtering, thereby retaining all reads at a given alignment start, and used the "-ERC GVCF" to produce per-sample genomic VCFs (gVCFs) containing both variant calls and reference-confidence blocks at every site which is information that is required for joint genotyping of the cohort in downstream steps. CombineGVCFs takes the intermediate gVCFs outputted by HaplotypeCaller and merges them into a single, combined gVCF to ensure that all per-sample annotations and confidence scores are preserved for efficient joint genotyping in the subsequent analysis step. Finally, GenotypeGVCFs was used on the combined gVCF file as input to perform joint genotyping. All samples are genotyped together, allowing the variant caller to more accurately distinguish true variants from sequencing error by considering how consistently a potential variant appears across the samples in the combined dataset. The output was a single gVCF that contained genotype calls for every sample.

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We visually inspected all microsatellite loci in the Integrative Genomic Viewer (IGV) (Robinson et al. 2011; Thorvaldsdóttir et al. 2013) to classify the loci as simple or complex. A simple locus had a single repeat array while a complex locus contained multiple or linked variants. As a check on genotyping accuracy, we spot-checked IGV-based classifications against PCR fragment-based and HaplotypeCaller genotypes, ensuring consistency between visual calls and our other methods of genotyping.

Phasing the gVCFs

To address discrepancies observed in the complex regions, specifically loci with multiple indels within the microsatellite, genotype phasing was performed. In these complex regions, multiple independent or linked indel events occur within the microsatellite, and treating each variant separately resulted in noticeable discrepancies between the PCR fragment size analysis results and the sequence-based genotype calls reported in the VCF files. To address this issue, the full combined, genotyped gVCF was phased using Beagle version 5.4, a software package for phasing genotypes, which organizes the alleles into distinct haplotypes so that variants occurring on the same chromosome can be linked (Browning et al. 2021). Phasing was performed on using the "excludemarkers" flag, which references a file listing markers to be excluded from the analysis. The regions in this file were NW_026622808.1:66387560, NW_026622863.1:14143646, NW_026622919.1:37502641, NW_026623078.1:38716578, NW_026623089.1:83246798, and NW_026623111.1:15116047. These excluded regions typically showed low complexity, low variation, or poor coverage that could interfere with accurate phasing. Removing these regions helped improve the overall reliability of the haplotype reconstruction, and the final phased output files reflect these exclusions.

Downsampling to different coverage depths

We implemented a downsampling strategy using GATK Picard DownsampleSam version 2.27.1 to simulate lower sequencing depths (McKenna et al. 2010). This tool randomly samples reads from BAM files to ensure that the process is unbiased and each read has an equal chance of being retained, thereby preserving the overall characteristics of the original dataset. To generate downsampled data sets at target coverages of 30x, 20x, 10x, 5x, and 2x, we employed the default usage of DownsampleSam so that it simply applies uniform random sampling to retain a fraction of the input reads. These coverages were chosen to represent a broad spectrum of typical sequencing depths from high coverage to low coverage.

The appropriate downsampling fraction was determined by calculating the average coverage of each BAM file using Samtools depth version 1.17 and then computing the ratio of the target coverage to the current, original coverage (Danecek et al. 2021). For example, if a BAM file originally had a coverage of 60x and our target downsampled coverage was 30x, the fraction would be 30/60 = 0.5, so that approximately half of the reads would be retained. Downsampled coverage was validated at each target depth (30x, 20x, 10x, 5x, 2x) in the same manner. These data allowed us to evaluate concordance across sample depths in order to determine the minimum depth required for reliable microsatellite genotyping. After downsampling, BAM files were processed as described above.

Concordance

Concordance between high coverage and PCR-based methods

To compare the output from GATK HaplotypeCaller with the PCR-based microsatellite typing data, we developed a series of custom scripts (https://github.com/dalyasalih19/Microsatellite_Comparison_Scripts) to compare high-coverage variant calls with PCR-based fragment length measurements.

The first script, pcr_data_excel_parsing.py, parses PCR allele length data from a CSV file in which each region is represented by two columns, one per allele. In this script, allele lengths are first sorted, and then a normalized genotype is derived by subtracting the smallest value from each allele's length (Fig. 1). This normalization is necessary because PCR-based genotyping reports the total fragments lengths including the flanking sequences, whereas sequencing-based tools report alleles as changes relative to the reference genome such as insertions, deletions, or differences in repeats. By comparing alleles based on their relative difference instead of their absolute fragment sizes, the normalization creates a consistent basis for comparison between the two methods.

A separate script, automated_hapcaller_comparisons.py, processes the HaplotypeCaller-generated VCF file. This script takes in a VCF and a set of predefined genomic regions as input and extracts all variant records within these regions. For each record, it constructs a temporary array that includes the length of the reference allele along with the lengths of the alternative alleles. By subtracting the smallest value from this array, the script converts the VCF genotype data into a set of normalized lengths analogous to the PCR-based genotypes. For complex regions, the script first determines whether more than one indel is present by scanning each VCF record within a user-defined region and checking if the length of any alternate allele differs from the reference allele—an indication of an insertion or deletion. Each record with at least one indel is counted once, and regions with more than one such record are classified as complex. For these complex regions, the script normalizes allele lengths by subtracting the shortest allele length across all reference and alternate alleles in a record. It then maps each sample's genotype to this normalized scale and sums these values across all indel records in the region. This summing step effectively captures the cumulative effect of all indel events across the entire PCR amplicon, a DNA fragment amplified using PCR, ensuring our PCR-based data were directly comparable with HaplotypeCaller's calls.

Concordance to high coverage versus downsampled coverages

To address the impact of sequencing depth on genotyping accuracy, we compared the high-coverage VCF data, which we used as the truth set, to the downsampled VCF data, which was used as the call set, generated using GATK Picard DownsampleSam at coverages 30x, 20x, 10x, 5x, and 2x.

We developed a custom Python script, diff_refs_genotype_concordance_hap_downsampling.py (https://github.com/dalyasalih19/Microsatellite_Comparison_Scripts), that processes each VCF file (truth and call sets) by parsing user-defined genomic regions and extracting all variant records within those regions. When the reference alleles in the datasets match, the script maps allele genotype codes (e.g.

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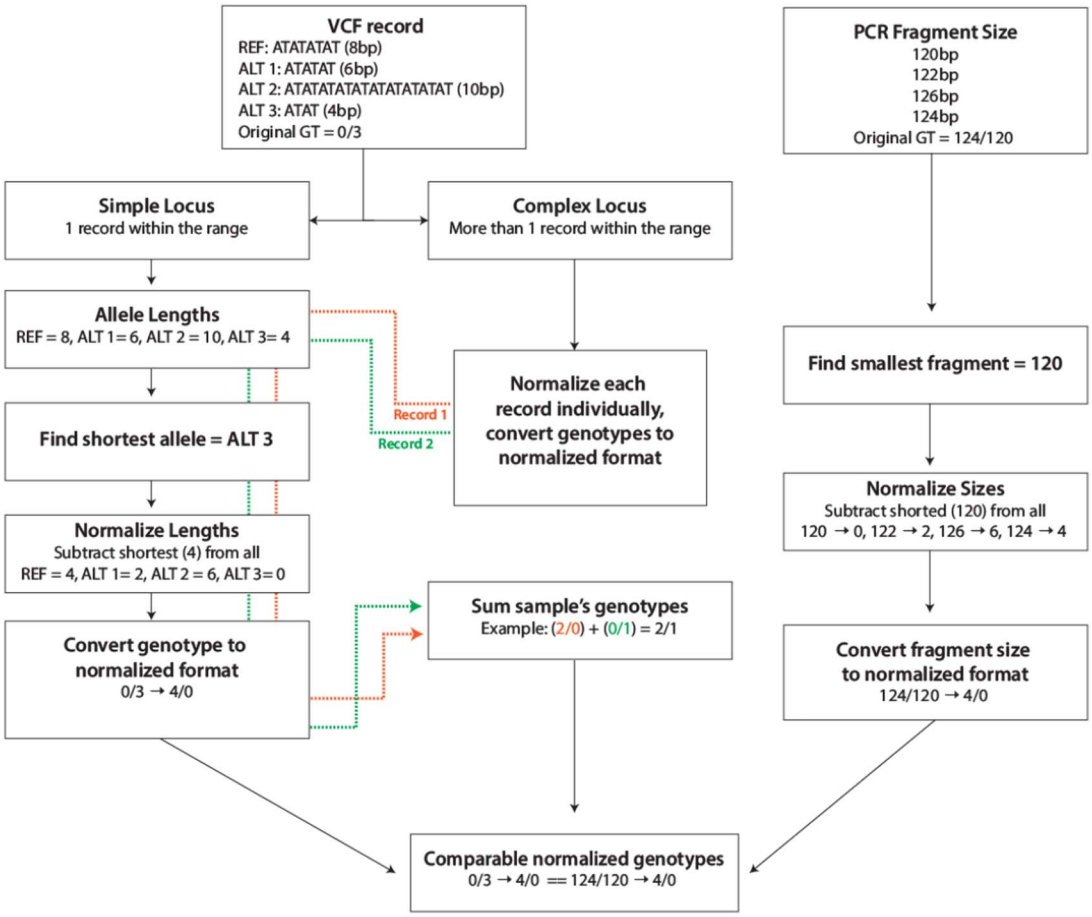


Fig. 1. Normalization and comparison of genotypes between HaplotypeCaller VCF records and PCR-based fragment data. VCF alleles are converted to bp lengths, normalized by subtracting the shortest allele, and mapped to genotypes. PCR fragment sizes are normalized in the same way. For complex loci, alleles are normalized per record and summed. This process places VCF- and PCR-derived genotypes on a shared scale for direct comparison.

0 for the reference and 1, 2, … for alternatives) to their corresponding nucleotide sequences, making the genotype conversion straightforward.

If the reference alleles differ between the high-coverage and downsampled datasets, the script computes an "allele difference" for each variant by subtracting the length of its reference allele from the allele's length; this quantifies the insertion or deletion size.

In such cases, we generate two mapping dictionaries for the downsampled (call) dataset: one using the downsampled reference as the baseline (with "0" set to 0) and a second with adjusted mapping where the reference allele is recalculated as the difference between the call and truth reference allele lengths. When converting genotypes, if a call genotype includes the reference allele or a "0" genotype code, the script chooses between the normal or adjusted mapping based on whether the truth genotype also includes a reference allele. This logic ensures that even when the call's reference allele corresponds to an alternate allele in the truth set, the downsampled genotype is mapped accurately to be directly comparable to the high-coverage data, which is critical for determining the minimum required sequencing depth for reliable genotyping. For the purposes of this study, genotypes are considered concordant only if both alleles exactly match the corresponding high-coverage genotype after conversion. Partial matches or ambiguous alleles are not counted as concordant.

Results

Microsatellites PCR-based typing

The 15 microsatellites were typed using duplicate runs and consensus genotypes were established (Table S2). Three alleles were confirmed in one individual at G10X, we classified the locus as a genotyping error and did not report any genotype for that sample. It was coded as NA/NA for comparison purposes.

Typing microsatellites in short-read data

Finding the microsatellites in the genome

Using the combined evidence of BLAST and repeat element information, we identified the location of commonly used microsatellite sequences in the high-quality brown bear reference genome assembly (Armstrong et al. 2022). In general, BLAST was enough to identify one clear, high-quality hit (Table S3). However, in several instances, searching the repetitive element database helped confirm the repeat's identity. For example, microsatellite Mu15 had three potential top hits that we followed up by searching for repetitive elements in these regions (Table S3). Though two of the three regions contained repetitive element sequences, one of these regions contained a simple repeat with a repeat motif that was only a length of one (i.e. a homopolymer), so we excluded this candidate region. The locations of each repeat in the reference genome

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Table 1. Concordance between genotypes called by GATK HaplotypeCaller and PCR-based methods across microsatellite loci. Loci are categorized as either simple (single variant, single location) or complex (linked or multiple variants). Concordance is reported as the fraction of matching genotypes out of total possible calls.

Microsatellite	Category (simple or complex)	Concordance
G10B	Simple	11/11 (100%)
G10C	Simple	11/11 (100%)
G10M	Simple	11/11 (100%)
G10P	Simple	11/11 (100%)
G10U	Simple	11/11 (100%)
G1A	Simple	11/11 (100%)
G1D	Simple	11/11 (100%)
Mu11_VIC	Simple	11/11 (100%)
Mu23	Simple	11/11 (100%)
Mu50	Simple	11/11 (100%)
REN145P07	Simple	11/11 (100%)
G10H	Complex	10/11 (91%)
G10L	Complex	11/11 (100%)
G10X	Complex	6/11 (55%)
Mu15	Complex	8/11 (73%)

are reported in Table S4. As expected, all of the microsatellites were dinucleotide repeats (Table S4).

Typing microsatellites with GATK HaplotypeCaller

We successfully completed all steps of the GATK HaplotypeCaller joint-calling pipeline and genotyped all 15 microsatellite loci of interest at full depth. The resulting gVCF was assessed using GATK's ValidateVariants tool, which returned no errors or warnings, indicating proper formatting and readiness for downstream analysis.

Concordance

Concordance between high coverage and PCR-based methods

Using our custom comparison scripts, we observed high genotype concordance between GATK HaplotypeCaller genotypes and those obtained through PCR-based methods. Of 165 possible genotypes, 156 matched exactly yielding a genotype concordance rate of 94.5% across the regions of interest. Lower genotype concordance occurred primarily in complex microsatellite regions, those with linked or multiple independent variants, where alignment and variant calling are more error-prone (see Table 1).

One locus, G10L, displayed a level of structural complexity that fragment-length analysis alone could not resolve. In this locus, two individuals carry two copies of an additional AC repeat unit, but the insertions occur at distinct positions on each allele within the same individual (Fig. 2). This created discordance when comparing genotyping approaches. PCR fragment size analysis captures the total amplicon length and the interpretation is that individuals are homozygous for a 2 bp insertion. However, HaplotypeCaller, which genotypes insertion or deletion (indel) positions independently at each nucleotide site in the region, identified two separate insertion sites, resulting in a heterozygous call for two different alternate alleles. This variation is invisible to fragment-based methods, which implicitly assumes that all size changes occur at a single site.

Another complex region where we encountered genotyping challenges was G10X, which contains a compound indel structure involving two separate deletion events occurring on the same haplotype. Specifically, one spans 5 bp at the start of the repeat (scaffold NW_026622763.1:42595152–42595157) and the other spans 4 bp just downstream of the repeat (scaffold NW_026622763.1: 42595216–42595219). Both deletions are observed together on the same sequencing reads in two individuals: K (100973) and P (100975). Because the indels appear together on individual reads, these mutations are physically linked (Fig. 3).

This kind of linkage presents a challenge for both traditional and modern genotyping methods. PCR fragment size analysis captures only the total amplicon size and would interpret G10X as a single 9 bp deletion, masking the fact that two distinct mutations are involved (see Fig. 3). While this simplifies data comparison, this interpretation loses important positional and structural information.

Another complex region was Mu15, which exhibited a linked sequence variant in individuals P (100975) and F (100977). On reads containing a TG insertion at position 22,888,533, there was also a consistent base substitution at position 22,888,574 (Fig. 4). HaplotypeCaller annotated this base change as an insertion event (G to GT), although manual inspection of the reads revealed a G > T substitution. This results in the sequence changing from TGGGGT to TGTGGT, effectively introducing two additional repeat units. Because these two variants are always found on the same reads, they appear to be physically linked. This complex structure likely contributed to the lower genotype concordance for Mu15 (Table 1), potentially due to differences in how variant types are interpreted by the calling algorithm.

Genotype concordance of high coverage versus downsampled datasets

To evaluate how sequencing depth affects genotype accuracy, we compared genotypes from downsampled BAM files (30x to 2x) to those from the original high-coverage data (32x to 62x). Actual depths are reported in Table S5. Genotype concordance was calculated between the highest coverage sample and the downsampled dataset.

At 30x coverage, 14 of the 15 microsatellite regions showed complete concordance between the highest coverage and the downsampled datasets, with only one region differing by a single sample genotype (Fig. 5). At 20x, 11 regions matched the original genotypes exactly; 3 regions had one discordant sample each, and 1 region had three discordant genotypes. At 10x, only 3 regions showed complete genotype concordance, with 3 others showing a single discordant sample each; the remaining 9 regions had reduced genotype concordance. At 5x coverage, concordance was generally low, with the highest agreement in any region being 81.82% meaning 9 out of 11 samples had matching genotypes. At 2x, concordance dropped dramatically. HaplotypeCaller failed to generate genotypes for 6 of the 15 loci and for the remaining 9 regions, the highest agreement in any of those regions was 45.45%, meaning only 5 of the 11 samples had concordant genotypes to the original coverage genotype.

Discussion

In this study, we compared traditional microsatellite genotype data with microsatellite genotypes derived from whole genome sequencing data in North American brown bears. We successfully located and genotyped 15 microsatellite

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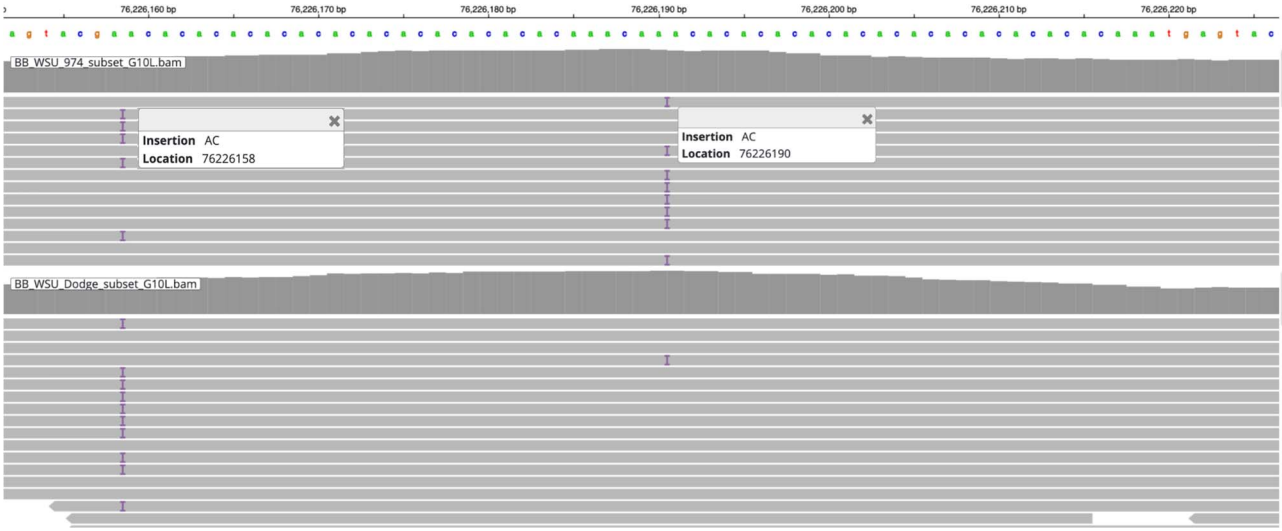


Fig. 2. IGV screenshot of the G10L region showing an extra copy of the AC repeat unit at two different genomic locations within the PCR region.



Fig. 3. IGV screenshot of the G10X region showing a complex haplotype with two linked deletions in individuals K (100973) and P (100975).

regions using GATK HaplotypeCaller's joint-calling pipeline. This approach allowed for sequence-resolved genotypes that captured both the size and structure of underlying repeat variation. Concordance between HaplotypeCaller and traditional PCR-based genotyping methods was generally high (Table 1), though some loci with complex repeat structures showed discordant results. This finding provides the important opportunity to link new whole genome datasets to historical microsatellite genotyping datasets for genetic monitoring of bears and other wildlife species (Schwartz et al. 2007). Genetic monitoring has increased rapidly as a tool for wildlife managers and allows researchers to estimate population size and density, track genetic variation, estimate effective population size, assess gene flow and connectivity, measure hybridization and introgression, and assess parentage and mating system (Stetz et al. 2011).

Repetitive regions are challenging to sequence and align reliably, which raises the possibility that discordant indel calls could result from alignment artifacts rather than true biological variation. To mitigate this, our pipeline incorporated joint genotyping across samples, which helps reduce false indel calls in low-complexity regions. Additionally, HaplotypeCaller includes built-in read filters for mapping quality which serve to discard reads with low alignment confidence. We used default settings to help reduce misaligned reads in repetitive regions. The specificity of the BLAST parameters and the lack of multiple candidate regions, along with the correspondence of the total microsatellite length derived from the BLAST search, also provided support for each microsatellite region being accurately identified. However, it is possible that the region could be misassembled in the reference, which we did not evaluate here. While no method can completely eliminate ambiguity in highly repetitive regions, the consistency of the observed allele-specific insertions across individuals supports that these represent true biological variation rather than misalignment artifacts. Additional long read genomes and resequencing data will help to verify these observations.

Understanding where repeat expansions and deletions occur matters for tracing mutational history, accurately reconstructing haplotypes, and ensuring consistent genotyping.

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Fig. 4. IGV screenshot of the Mu15 region showing a complex haplotype with a linked insertion and single-nucleotide polymorphism (SNP) in individuals P (100975) and F (100977).

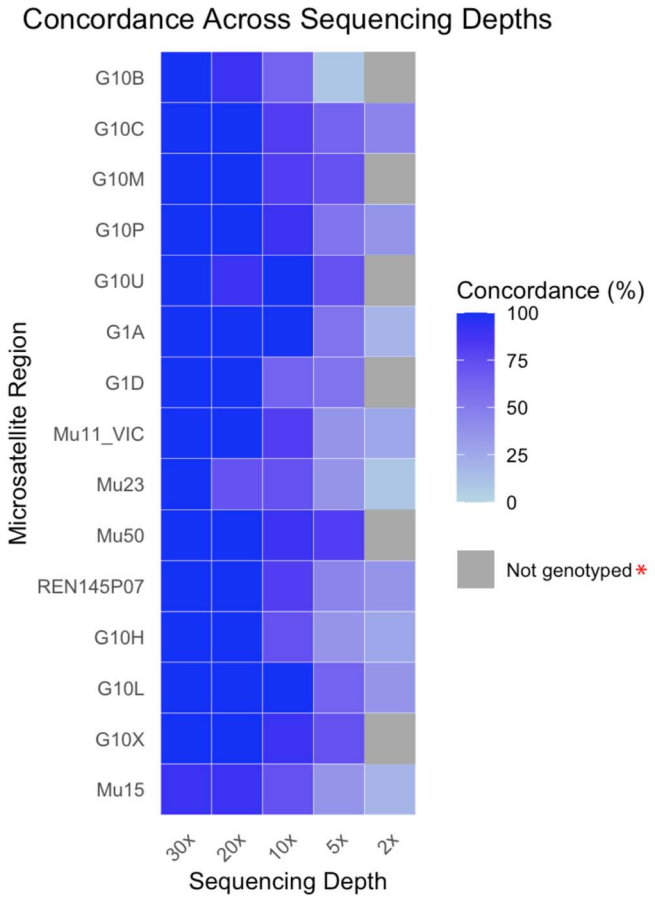


Fig. 5. Genotype concordance across sequencing depths for 15 microsatellite regions. Rows represent individual loci; columns represent sequencing depths from 30x to 2x. Colors indicate depth concordance. For actual values, see Table S6.

Each insertion may reflect an independent mutational event, and identifying their locations allows us to capture the locus's evolutionary history more accurately (Ellegren 2004). HaplotypeCaller's variant calls preserve this structure and reveal heterogeneity that fragment data alone cannot. This is especially critical when aligning legacy microsatellite data with modern sequencing-based genotypes: multiple distinct alleles can result in the same PCR length, but have very different biological and evolutionary implications.

For instance, in G10L, two individuals carried additional AC repeat units at different positions on each allele. While PCR fragment size analysis measured identical amplicon sizes and reported homozygosity, HaplotypeCaller detected insertions at distinct positions, resulting in heterozygous genotypes. This phenomenon, known as homoplasy, occurs when independent mutational events converge on identical fragment sizes. This discrepancy reflects a fundamental difference in how these methods interpret variation. PCR captures length alone, while sequence-based tools reveal positional structure (Fig. 2).

A more complex case occurred in G10X, where two deletions on the same haplotype were misclassified as independent events. In sample K (100973), this resulted in an implausible genotype with three separate variant calls: two deletions and one insertion. In reality, the individual carries one haplotype with a compound deletion and another with a simple insertion (Fig. 3). Because HaplotypeCaller lacked phasing across the entire region, it failed to recognize the linkage between the deletions. Even with the additional phasing we incorporated, HaplotypeCaller still failed to link the two deletions, likely because very few individual reads spanned the full extent of the repetitive region, preventing it from recognizing the true haplotype of the complex region.

A third case, Mu15, illustrates how sequence context can complicate variant interpretation. In this locus, a TG insertion at NW_026623019.1:2,888,533 consistently co-occurs with a G-T substitution at NW_026623019.1:2,888,574. The resulting sequence changes from TGGGGT to TGTGGT. HaplotypeCaller misclassified the substitution as an insertion from G to GT (Fig. 4) while PCR would likely not detect this variant at all, since the substitution does not change the total amplicon length. In contrast, sequence-based tools may annotate it as an

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insertion or treat it as a SNP within a repeat, contributing to discordance.

To address the challenges created by homoplasy, we developed custom Python-based scripts that scans each target region and sums indel variants per sample when multiple indels are present. Each approach, on its own, provides only a partial picture. PCR simplifies complex alleles into single size differences, while variant calling without phasing may overcomplicate or mislabel them (Lepais et al. 2020). The tools developed here make it possible to compare PCR length data with sequence-based genotypes in a consistent and interpretable way, overcoming each method's limitations. By using sequence-based STR tools, we can uncover hidden heterogeneity within loci that would otherwise appear uniform, leading to more accurate genotyping and a clearer understanding of repeat dynamics.

However, variant callers for short-read sequence data such as GATK HaplotypeCaller (McKenna et al. 2010) analyze each insertion or deletion independently, which can artificially split what is biologically a single compound haplotype into multiple distinct variants. Left uncorrected, this over-fragmentation can create genotypes that are implausible, obscure the true allele structure, and inflate apparent allele counts. Our aggregation strategy resolves this by grouping linked variants back into coherent haplotypes, ensuring that true biological heterogeneity is preserved while avoiding artifacts of oversimplification or over-fragmentation. For example, in complex loci with multiple indels, PCR might report a uniform 2 bp insertion, while HaplotypeCaller might call two separate single-base insertions; aggregating these restores a consistent representation of the haplotype while still retaining the biologically meaningful distinction that PCR alone would have missed.

A similar bridge between legacy capillary data and full sequence resolution was demonstrated by (Vartia et al. 2016), who used combinatorial PCR barcoding and high-throughput sequencing to genotype 53 Atlantic cod microsatellites. Their method, like ours, recovered both repeat lengths and internal sequence variation, enabling the detection of homoplasy and positional variants that fragment size analysis would miss. However, their approach was species-specific and based on a fixed marker panel, whereas our method is designed to be more flexible across species and loci. Šarhanová et al. (2018) also compared traditional fragment-length scoring with sequencing-based genotyping using SSR-seq, showing that sequence data revealed a higher level of polymorphism and allowed more accurate identification of alleles. Our work builds on these studies by focusing not only on the improved resolution offered by sequencing but also on how sequence-based data can be aligned with and interpreted alongside existing fragment-based datasets.

Correctly resolving complex alleles depends on phasing. Without physical phasing, insertions may be misinterpreted as separate indels, compound alleles, or missed altogether. When reads do not span the full locus or when evidence is limited to a small number of individuals, HaplotypeCaller and similar tools may fail to link multiple variants together. In G10X, for example, few reads span the full repeat and its flanking regions, and only 2 of the 11 individuals carry the complex structure (Fig. 3). This lack of physical phasing data and limited support reduces the tool's ability to assign variants to the correct haplotypes.

To validate the accuracy of STR-genotyping at various sequencing depths, we downsampled high coverage BAM files and re-ran the HaplotypeCaller joint-calling pipeline. We then compared genotype calls from the downsampled datasets to those from the original full-depth run using a custom Python script. As sequencing depth decreased, we observed a corresponding decline in genotype concordance. At higher coverage depths, from 20 to 30x, there was high concordance. However, the results for the 10 to 2x highlighted several issues introduced by low coverage. First, insufficient read depth prevented confident genotyping all together, as seen at 2x where entire regions were missed (Fig. 5; Table S6). Second, when genotypes were called, they were often driven by single-reads and were poorly supported. With low-coverage datasets, the likelihood of finding complex variants significantly decreases due to insufficient read coverage. Importantly, we found that homozygous genotypes tended to show higher concordance than heterozygous ones, likely because low depth makes it harder to detect reads supporting both alleles to make a confident heterozygous call, emphasizing the need of having adequate read coverage.

Even at high coverage, certain regions remained difficult to genotype accurately. In particular, loci with long or repetitive sequences often had fewer reads that spanned the full region from end to end. Without these full-length reads, it becomes difficult to reconstruct the correct haplotypes or resolve multiple linked variants emphasizing the need of having the right type of read coverage. Additionally, genotyping at depths below 20 to 30x led to missed calls when coverage was too sparse for accurate variant detection. Lower sequencing depths resulted in poorly supported calls based on few or single reads. While recent methods have improved genotype calling from low-coverage sequencing data using imputation or optimized algorithms (e.g. (Zan et al. 2019, Zhang et al. 2025)), and targeted sequencing strategies such as amplicon-based genotyping or hybrid capture can achieve high local coverage with fewer reads, our analysis was based on whole-genome sequencing without the use of imputation. As such, our depth recommendations reflect performance under those conditions. Our inference of these regions would also likely be improved with long read data, but this is unlikely to be feasible for many low-quality samples used for wildlife monitoring.

Lastly, complex regions posed challenges even at higher coverage, and lowering depth further reduced the ability to reconstruct compound haplotypes without reads spanning all linked sites. To mitigate these issues, we recommend a minimum sequencing depth ranging from 20 to 30x for STR genotyping using this pipeline. Furthermore, we recommend caution if including complex loci in comparative analyses. These regions consistently showed low genotyping concordance and a greater difficulty in calling accurate genotypes. Instead, focusing on simple loci led to more reliable comparisons and biologically meaningful conclusions, where coverage was sufficient, alleles were more clearly resolved and concordance with legacy PCR-based data was higher.

This study is the first to resolve a critical problem in genetic monitoring of wildlife, which is standardizing historic microsatellite datasets to data generated with new sequencing technologies. Many researchers have not transitioned to high throughput sequencing due to the high costs and time associated with regenotyping/sequencing all samples in long-term monitoring databases. This technique provides a low-

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cost option for standardizing on a few (10 to 20) samples as an approach to convert a full database. Overall, this will facilitate the transition from PCR based-fragment size analysis genotyping to whole genome sequencing of wildlife samples. This approach will be most successful when working with high quality DNA sources like blood or tissue and is less likely to be successful when working with low quantity DNA sources such as hair and feces. There are also challenges for laboratories to develop the bioinformatic processing skills necessary to transition to high-throughput sequencing, but the methods and scripts provided here should be helpful in that transition. Another benefit of this approach for genetic monitoring is that high-throughput sequencing will generate data that is compatible with long-term microsatellite datasets but at the same time generates results for millions of neutral and potentially adaptive loci while microsatellite loci are generally neutral and limited in number. This greatly expands the number and types of questions that can be addressed with genetic monitoring.

Conclusion

In this study, we evaluated the performance of short-read whole-genome sequencing for microsatellite genotyping and compared it with traditional PCR-based methods. Microsatellites remain a valuable genetic tool, especially in conservation research where decades of data have been generated using PCR-based genotyping (Allendorf et al. 2022). As sequencing technologies advance, it becomes increasingly important to ensure that new methods can align with and extend the value of these legacy datasets. Using GATK HaplotypeCaller, we successfully genotyped 15 microsatellite loci in brown bears and achieved high concordance with PCR data, especially in simple regions, namely those with a single indel within the microsatellite region. However, we also found that complex loci, those containing multiple or linked variants, present challenges for both PCR and standard variant calling tools. Standard variant calling tools often fragmented compound haplotypes into separate indels, while PCR failed to resolve positional differences within alleles of identical length.

To address these challenges, we developed custom scripts that aggregate indels within a locus, enabling direct comparison between sequence-based calls and PCR fragment data. This approach recovered more accurate genotypes and clarified discrepancies in complex regions such as G10L, G10X, and Mu15. Downsampling analyses revealed that genotyping accuracy drops significantly at low coverage. Below 10x, many loci failed entirely or yielded unreliable results. We recommend 20 to 30x coverage for reliable STR genotyping with this pipeline. However, depth alone is not sufficient, reads should also span the full microsatellite and its flanking sequence to resolve haplotypes correctly. Therefore, longer reads should be used for longer microsatellite regions. Our results demonstrate that legacy PCR-based microsatellite data can be meaningfully aligned with modern high-throughput sequencing. By refining comparison strategies and understanding the limitations of each method, we can better preserve historical data, improve genotyping accuracy, and support future genetic research. This integration offers a practical strategy for extending the utility of historical datasets while improving resolution and accuracy in contemporary conservation genetics.

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Author contributions

Dalya Salih (Data curation, Formal analysis, Methodology, Software, Visualization, Writing—original draft, Writing—review & editing), Ellie Armstrong (Data curation, Investigation, Writing—review & editing), Charles Robbins (Resources, Writing—review & editing), Lisette Waits (Formal analysis, Investigation, Writing—review & editing), and Joanna Kelley (Conceptualization, Methodology, Project administration, Resources, Supervision, Writing—original draft, Writing—review & editing)

Supplementary material

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Data availability

Raw DNA sequence reads are available under SRA BioProject PRJNA613079.

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